

CARGOMULE · REQUIREMENTS

CargoMule requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	CGM-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/cargomule		
Source	docs/03-requirements.md @ d8100b5		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

CargoMule requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3. Status is judged against the first-order estimates in CGM-PRC-001; **four requirements are not met or at risk** (R7, R8, R12 and the thermal part of R3).

The **design load case** is 150 kg of payload centered on the deck, a trailer of about 38 kg (estimate) and a towing bicycle with rider of about 100 kg, on a dry paved road. The **design route** is a 10 km round trip with 120 m of total climbing and 20 stops.

Table 1. Requirements, targets and concept status.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Carry the payload	150 kg (330 lb) on a deck of at least 0.8 m ² , with tie-down points	Frame stress calculation; later proof load	Deck 0.84 m ² ; frame strength unverified
R2	Range per charge	20 km or more on the design route in the design load case	Energy calculation; later logged ride	Met: about 24 km (estimate)
R3	Assist on hills	8 % grade for 300 m at 8 km/h or more with the rider feeling 40 N or less of drawbar pull, without motor over-temperature at 30 °C	Force and thermal calculation	Force met (about 33 N); thermal unverified , motor near 290 W against a 250 W rating
R4	Near-zero added load on the flat	Steady drawbar pull felt by the rider 15 N or less at 5 to 20 km/h	Control calculation and simulation	Met on paper (about 4 N); loop stability unverified
R5	Stay within pedal-assist limits	250 W rated continuous motor output; assist only while the drawbar is in tension; no assist above 25 km/h; no throttle	Datasheets; controller settings	Met by design
R6	Trailer brakes itself	With the combination slowing at 3 m/s ² , the trailer pushes the bicycle with 100 N or less; assist cut within 100 ms of drawbar compression above 20 N	Braking calculation; later bench test	By design (overrun brake); gain unverified
R7	Hitch to common bicycles	Fits rear wheels with a 9 mm quick release or a 12 mm thru-axle, 130 to 148 mm spacing, 26 in to 700c; hitch and unhitch in 30 s or less; no wiring to the bike	Hitch design review; later fit survey	At risk: nutted axles, some hub gears and some rear-motor e-bikes need adapters, so "any bicycle" is not met
R8	Light enough to handle	Empty trailer 35 kg (77 lb) or less, including the pack	Mass estimate; later weighing	Not met: about 38 kg with a steel frame (estimate)
R9	Fit bike paths and doors	Overall width 1,000 mm or less; length 2.6 m or less hitched from the bike's rear axle	Model check	Met: about 910 mm wide, about 2.5 m long
R10	Stable hitch load	Downward load on the hitch 3 to 10 kg with the design payload centered	Mass and balance estimate	Met: about 9 kg (estimate); off-center loads unverified
R11	Weather and temperature	Electronics IP65; motor IP54 or better; operate –10 to 40 °C; the BMS blocks charging below 0 °C	Datasheets and design review	By design; unverified
R12	Affordable	Prototype parts \$900 or less, including charger	Priced BOM (bom/bom.csv)	Not met: about \$965 (indicative), about 7 % over
R13	Fail safe	Loss of the load cell signal, a reading out of range, a stuck value or a controller fault sets motor current to zero within 100 ms; the trailer stays towable as an unpowered, braked trailer	Fault tree and design review	By design; unverified
R14	Be seen		Design review	Met by design

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
		Rear red lights and reflectors, side reflectors, and a flag at 1.5 m or more above the ground		

■ Assumptions

- Rolling resistance coefficient 0.010 for 20 x 2.15 in tyres on pavement at design load; extra drag area of the loaded trailer 0.1 m², since it rides partly in the rider's wake.
- Air density 1.2 kg/m³; gravitational acceleration 9.81 m/s².
- Motor, gear and controller efficiency 75 %; charger efficiency 90 %; cell charging efficiency 96 % for LiFePO₄.
- Usable pack energy 90 % of the nominal 384 Wh.
- 250 W and 25 km/h follow the EU pedal-assist exclusion (Regulation (EU) No 168/2013); the US federal definition allows more (under 750 W, 20 mph), so the EU limits are the tighter case.
- Brake and hitch loads follow the voluntary cycle trailer standards EN 15918 and ASTM F1975, which limit unbraked trailers to 60 kg and 45.4 kg; CargoMule exceeds both, so its brakes are required, not optional.